

V3 parameter estimation

Seven models · SPY, AAPL, MSFT, AMZN · four windows · 1.5-year fixed calibration

What this report is

- The single t_0 parameter vector used in this study, summarized by regime. One table per model. $n = 1$ so SD is —.
- No LSM, no option prices, and no simulated paths. Only calibrate_ticker with the 18-month lookback.
- The companion notebook is the computational source of truth. This PDF is written from the same payload.

How estimates are formed

- At the first session t_0 of each 1-year evaluation window, parameters use only returns and listed quotes with `trading_date ≤ t0`. Those parameters are held for every Monday contract in the window. LSM prices each contract with that single calibration row. Option quotes are as-of the grid timestamp (same session, else the prior session).
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t_0 are used only after t_0 .
- Cells are Mean, SD, Min, and Max. With $n = 1$, Mean = Min = Max and SD is —.

Tables

- P1 GBM. P2 Modified GBM. P3 GARCH. P4 Heston. P5 Merton. P6 GARCH-Merton. P7 Heston-Merton.
- R1 Crisis, R2 Normal, R3 Late-cycle, R4 COVID. Columns are SPY, AAPL, MSFT, AMZN.

Table P1 · GBM · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

gime / para	SPY				AAPL				MSFT				AMZN			
gime / para	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ	-0.0912	—	-0.0912	-0.0912	0.4097	—	0.4097	0.4097	-0.1082	—	-0.1082	-0.1082	0.4477	—	0.4477	0.4477
σ	0.1852	—	0.1852	0.1852	0.4044	—	0.4044	0.4044	0.2730	—	0.2730	0.2730	0.5045	—	0.5045	0.5045
R2 Normal																
μ	0.1945	—	0.1945	0.1945	-0.0234	—	-0.0234	-0.0234	0.1597	—	0.1597	0.1597	0.3665	—	0.3665	0.3665
σ	0.1147	—	0.1147	0.1147	0.2892	—	0.2892	0.2892	0.2339	—	0.2339	0.2339	0.2668	—	0.2668	0.2668
R3 Late-cycle																
μ	0.1419	—	0.1419	0.1419	0.3211	—	0.3211	0.3211	0.3944	—	0.3944	0.3944	0.5438	—	0.5438	0.5438
σ	0.1100	—	0.1100	0.1100	0.2073	—	0.2073	0.2073	0.2007	—	0.2007	0.2007	0.2412	—	0.2412	0.2412
R4 COVID																
μ	0.0514	—	0.0514	0.0514	0.1183	—	0.1183	0.1183	0.2690	—	0.2690	0.2690	0.1177	—	0.1177	0.1177
σ	0.1544	—	0.1544	0.1544	0.2924	—	0.2924	0.2924	0.2562	—	0.2562	0.2562	0.3257	—	0.3257	0.3257

Notes: This table reports Mean, SD, Min, and Max of the none GBM estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ , σ . Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P2 · Modified GBM · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
P(U U)	0.4602	—	0.4602	0.4602	0.5071	—	0.5071	0.5071	0.4918	—	0.4918	0.4918	0.4335	—	0.4335	0.4335
P(D D)	0.3961	—	0.3961	0.3961	0.3899	—	0.3899	0.3899	0.5079	—	0.5079	0.5079	0.4395	—	0.4395	0.4395
P(U D)	0.6039	—	0.6039	0.6039	0.6101	—	0.6101	0.6101	0.4921	—	0.4921	0.4921	0.5605	—	0.5605	0.5605
P(D U)	0.5398	—	0.5398	0.5398	0.4929	—	0.4929	0.4929	0.5082	—	0.5082	0.5082	0.5665	—	0.5665	0.5665
μ_U	0.0077	—	0.0077	0.0077	0.0190	—	0.0190	0.0190	0.0127	—	0.0127	0.0127	0.0230	—	0.0230	0.0230
σ_U	0.0073	—	0.0073	0.0073	0.0148	—	0.0148	0.0148	0.0118	—	0.0118	0.0118	0.0293	—	0.0293	0.0293
μ_D	0.0095	—	0.0095	0.0095	0.0200	—	0.0200	0.0200	0.0131	—	0.0131	0.0131	0.0191	—	0.0191	0.0191
σ_D	0.0085	—	0.0085	0.0085	0.0185	—	0.0185	0.0185	0.0113	—	0.0113	0.0113	0.0171	—	0.0171	0.0171
R2 Normal																
P(U U)	0.5678	—	0.5678	0.5678	0.4572	—	0.4572	0.4572	0.4921	—	0.4921	0.4921	0.4974	—	0.4974	0.4974
P(D D)	0.4488	—	0.4488	0.4488	0.4738	—	0.4738	0.4738	0.4724	—	0.4724	0.4724	0.4947	—	0.4947	0.4947
P(U D)	0.5512	—	0.5512	0.5512	0.5262	—	0.5262	0.5262	0.5276	—	0.5276	0.5276	0.5053	—	0.5053	0.5053
P(D U)	0.4322	—	0.4322	0.4322	0.5428	—	0.5428	0.5428	0.5079	—	0.5079	0.5079	0.5026	—	0.5026	0.5026
μ_U	0.0055	—	0.0055	0.0055	0.0132	—	0.0132	0.0132	0.0107	—	0.0107	0.0107	0.0139	—	0.0139	0.0139
σ_U	0.0047	—	0.0047	0.0047	0.0110	—	0.0110	0.0110	0.0101	—	0.0101	0.0101	0.0125	—	0.0125	0.0125
μ_D	0.0054	—	0.0054	0.0054	0.0131	—	0.0131	0.0131	0.0099	—	0.0099	0.0099	0.0111	—	0.0111	0.0111
σ_D	0.0049	—	0.0049	0.0049	0.0142	—	0.0142	0.0142	0.0114	—	0.0114	0.0114	0.0100	—	0.0100	0.0100
R3 Late-cy																
P(U U)	0.5532	—	0.5532	0.5532	0.5371	—	0.5371	0.5371	0.5233	—	0.5233	0.5233	0.6149	—	0.6149	0.6149
P(D D)	0.4018	—	0.4018	0.4018	0.4688	—	0.4688	0.4688	0.3531	—	0.3531	0.3531	0.4554	—	0.4554	0.4554
P(U D)	0.5982	—	0.5982	0.5982	0.5312	—	0.5312	0.5312	0.6469	—	0.6469	0.6469	0.5446	—	0.5446	0.5446
P(D U)	0.4468	—	0.4468	0.4468	0.4629	—	0.4629	0.4629	0.4767	—	0.4767	0.4767	0.3851	—	0.3851	0.3851
μ_U	0.0045	—	0.0045	0.0045	0.0100	—	0.0100	0.0100	0.0092	—	0.0092	0.0092	0.0110	—	0.0110	0.0110
σ_U	0.0042	—	0.0042	0.0042	0.0095	—	0.0095	0.0095	0.0090	—	0.0090	0.0090	0.0112	—	0.0112	0.0112
μ_D	0.0047	—	0.0047	0.0047	0.0088	—	0.0088	0.0088	0.0087	—	0.0087	0.0087	0.0104	—	0.0104	0.0104
σ_D	0.0063	—	0.0063	0.0063	0.0086	—	0.0086	0.0086	0.0093	—	0.0093	0.0093	0.0106	—	0.0106	0.0106
R4 COVID																
P(U U)	0.5585	—	0.5585	0.5585	0.5515	—	0.5515	0.5515	0.5186	—	0.5186	0.5186	0.5738	—	0.5738	0.5738
P(D D)	0.4856	—	0.4856	0.4856	0.4829	—	0.4829	0.4829	0.3673	—	0.3673	0.3673	0.4673	—	0.4673	0.4673
P(U D)	0.5144	—	0.5144	0.5144	0.5171	—	0.5171	0.5171	0.6327	—	0.6327	0.6327	0.5327	—	0.5327	0.5327
P(D U)	0.4415	—	0.4415	0.4415	0.4485	—	0.4485	0.4485	0.4814	—	0.4814	0.4814	0.4262	—	0.4262	0.4262
μ_U	0.0065	—	0.0065	0.0065	0.0125	—	0.0125	0.0125	0.0112	—	0.0112	0.0112	0.0134	—	0.0134	0.0134
σ_U	0.0061	—	0.0061	0.0061	0.0117	—	0.0117	0.0117	0.0107	—	0.0107	0.0107	0.0135	—	0.0135	0.0135
μ_D	0.0072	—	0.0072	0.0072	0.0135	—	0.0135	0.0135	0.0124	—	0.0124	0.0124	0.0157	—	0.0157	0.0157
σ_D	0.0078	—	0.0078	0.0078	0.0146	—	0.0146	0.0146	0.0118	—	0.0118	0.0118	0.0159	—	0.0159	0.0159

Notes: This table reports Mean, SD, Min, and Max of the none Modified GBM estimates for SPY, AAPL, MSFT, AMZN. Parameters are P(U|U), P(D|D), P(U|D), P(D|U), μ_U, σ_U, μ_D, σ_D. Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P3 · GARCH · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

gime / para	SPY				AAPL				MSFT				AMZN			
gime / para	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ	-0.0305	—	-0.0305	-0.0305	0.6989	—	0.6989	0.6989	-0.0817	—	-0.0817	-0.0817	0.5326	—	0.5326	0.5326
ω	2.695e-06	—	2.695e-06	2.695e-06	4.710e-05	—	4.710e-05	4.710e-05	1.204e-05	—	1.204e-05	1.204e-05	9.001e-04	—	9.001e-04	9.001e-04
α	0.0554	—	0.0554	0.0554	0.0987	—	0.0987	0.0987	0.1011	—	0.1011	0.1011	0.1101	—	0.1101	0.1101
β	0.9275	—	0.9275	0.9275	0.8310	—	0.8310	0.8310	0.8678	—	0.8678	0.8678	3.061e-14	—	3.061e-14	3.061e-14
σ ₀	0.0140	—	0.0140	0.0140	0.0262	—	0.0262	0.0262	0.0238	—	0.0238	0.0238	0.0312	—	0.0312	0.0312
R2 Normal																
μ	0.2315	—	0.2315	0.2315	0.1421	—	0.1421	0.1421	0.2948	—	0.2948	0.2948	0.4583	—	0.4583	0.4583
ω	1.379e-05	—	1.379e-05	1.379e-05	6.339e-06	—	6.339e-06	6.339e-06	1.340e-04	—	1.340e-04	1.340e-04	4.158e-05	—	4.158e-05	4.158e-05
α	0.1358	—	0.1358	0.1358	0.0382	—	0.0382	0.0382	0.1734	—	0.1734	0.1734	0.1263	—	0.1263	0.1263
β	0.6047	—	0.6047	0.6047	0.9443	—	0.9443	0.9443	0.2481	—	0.2481	0.2481	0.7358	—	0.7358	0.7358
σ ₀	0.0061	—	0.0061	0.0061	0.0151	—	0.0151	0.0151	0.0134	—	0.0134	0.0134	0.0154	—	0.0154	0.0154
R3 Late-cycle																
μ	0.2143	—	0.2143	0.2143	0.3426	—	0.3426	0.3426	0.4942	—	0.4942	0.4942	0.7010	—	0.7010	0.7010
ω	2.969e-06	—	2.969e-06	2.969e-06	4.492e-05	—	4.492e-05	4.492e-05	4.208e-05	—	4.208e-05	4.208e-05	5.971e-05	—	5.971e-05	5.971e-05
α	0.2082	—	0.2082	0.2082	0.1209	—	0.1209	0.1209	0.2060	—	0.2060	0.2060	0.1350	—	0.1350	0.1350
β	0.7275	—	0.7275	0.7275	0.6152	—	0.6152	0.6152	0.5131	—	0.5131	0.5131	0.6153	—	0.6153	0.6153
σ ₀	0.0044	—	0.0044	0.0044	0.0125	—	0.0125	0.0125	0.0096	—	0.0096	0.0096	0.0143	—	0.0143	0.0143
R4 COVID																
μ	0.2136	—	0.2136	0.2136	0.5268	—	0.5268	0.5268	0.3807	—	0.3807	0.3807	0.5059	—	0.5059	0.5059
ω	4.718e-06	—	4.718e-06	4.718e-06	1.819e-05	—	1.819e-05	1.819e-05	2.013e-05	—	2.013e-05	2.013e-05	1.517e-05	—	1.517e-05	1.517e-05
α	0.1684	—	0.1684	0.1684	0.1773	—	0.1773	0.1773	0.1825	—	0.1825	0.1825	0.2090	—	0.2090	0.2090
β	0.7886	—	0.7886	0.7886	0.7917	—	0.7917	0.7917	0.7474	—	0.7474	0.7474	0.7677	—	0.7677	0.7677
σ ₀	0.0107	—	0.0107	0.0107	0.0186	—	0.0186	0.0186	0.0143	—	0.0143	0.0143	0.0142	—	0.0142	0.0142

Notes: This table reports Mean, SD, Min, and Max of the none GARCH estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ , ω , α , β , σ_0 . Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P4 · Heston · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (2008)																
μ	-0.0912	—	-0.0912	-0.0912	0.4097	—	0.4097	0.4097	-0.1082	—	-0.1082	-0.1082	0.4477	—	0.4477	0.4477
κ	5.7004	—	5.7004	5.7004	17.0428	—	17.0428	17.0428	3.9891	—	3.9891	3.9891	9.7392	—	9.7392	9.7392
θ	0.0523	—	0.0523	0.0523	0.1808	—	0.1808	0.1808	1.001e-04	—	1.001e-04	1.001e-04	0.2647	—	0.2647	0.2647
ξ	0.8445	—	0.8445	0.8445	2.5670	—	2.5670	2.5670	0.5021	—	0.5021	0.5021	2.3764	—	2.3764	2.3764
ρ	-0.7055	—	-0.7055	-0.7055	-0.4119	—	-0.4119	-0.4119	-0.3555	—	-0.3555	-0.3555	-0.5077	—	-0.5077	-0.5077
v_0	0.0412	—	0.0412	0.0412	0.1779	—	0.1779	0.1779	0.1117	—	0.1117	0.1117	0.2444	—	0.2444	0.2444
R2 Normal (2009-2014)																
μ	0.1945	—	0.1945	0.1945	-0.0234	—	-0.0234	-0.0234	0.1597	—	0.1597	0.1597	0.3665	—	0.3665	0.3665
κ	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	1.9081	—	1.9081	1.9081
θ	0.0248	—	0.0248	0.0248	0.1227	—	0.1227	0.1227	0.0637	—	0.0637	0.0637	1.0296	—	1.0296	1.0296
ξ	2.3743	—	2.3743	2.3743	1.8456	—	1.8456	1.8456	0.0833	—	0.0833	0.0833	3.9998	—	3.9998	3.9998
ρ	-0.6720	—	-0.6720	-0.6720	0.0453	—	0.0453	0.0453	0.9900	—	0.9900	0.9900	-0.2458	—	-0.2458	-0.2458
v_0	0.0200	—	0.0200	0.0200	0.0435	—	0.0435	0.0435	0.0411	—	0.0411	0.0411	0.0689	—	0.0689	0.0689
R3 Late-cycle (2015-2018)																
μ	0.1419	—	0.1419	0.1419	0.3211	—	0.3211	0.3211	0.3944	—	0.3944	0.3944	0.5438	—	0.5438	0.5438
κ	14.7216	—	14.7216	14.7216	12.0825	—	12.0825	12.0825	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000
θ	0.0196	—	0.0196	0.0196	0.1392	—	0.1392	0.1392	0.0778	—	0.0778	0.0778	0.2211	—	0.2211	0.2211
ξ	1.6499	—	1.6499	1.6499	3.2057	—	3.2057	3.2057	1.3339	—	1.3339	1.3339	3.9592	—	3.9592	3.9592
ρ	-0.6219	—	-0.6219	-0.6219	-0.4728	—	-0.4728	-0.4728	-0.4759	—	-0.4759	-0.4759	-0.4511	—	-0.4511	-0.4511
v_0	0.0078	—	0.0078	0.0078	0.0366	—	0.0366	0.0366	0.0176	—	0.0176	0.0176	0.0370	—	0.0370	0.0370
R4 COVID (2020)																
μ	0.0514	—	0.0514	0.0514	0.1183	—	0.1183	0.1183	0.2690	—	0.2690	0.2690	0.1177	—	0.1177	0.1177
κ	20.0000	—	20.0000	20.0000	19.1300	—	19.1300	19.1300	16.6505	—	16.6505	16.6505	19.5757	—	19.5757	19.5757
θ	0.0326	—	0.0326	0.0326	0.1061	—	0.1061	0.1061	0.0797	—	0.0797	0.0797	0.0955	—	0.0955	0.0955
ξ	2.2686	—	2.2686	2.2686	2.2326	—	2.2326	2.2326	2.2571	—	2.2571	2.2571	1.4660	—	1.4660	1.4660
ρ	-0.8074	—	-0.8074	-0.8074	-0.6065	—	-0.6065	-0.6065	-0.5863	—	-0.5863	-0.5863	-0.7811	—	-0.7811	-0.7811
v_0	0.0500	—	0.0500	0.0500	0.1016	—	0.1016	0.1016	0.0767	—	0.0767	0.0767	0.0572	—	0.0572	0.0572

Notes: This table reports Mean, SD, Min, and Max of the none Heston estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ , κ , θ , ξ , ρ , v_0 . Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P5 · Merton · none parameter estimates

1.5-year lookahead · rolling=none · Mean, SD, Min, Max within each regime

Regime / para	SPY				AAPL				MSFT				AMZN			
Regime / para	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (																
μ	-0.1447	—	-0.1447	-0.1447	0.4746	—	0.4746	0.4746	-0.0396	—	-0.0396	-0.0396	0.0224	—	0.0224	0.0224
σ	0.1725	—	0.1725	0.1725	0.3680	—	0.3680	0.3680	0.2379	—	0.2379	0.2379	0.3758	—	0.3758	0.3758
λ	2.6667	—	2.6667	2.6667	2.0000	—	2.0000	2.0000	2.6667	—	2.6667	2.6667	4.0000	—	4.0000	4.0000
$\mu_{\downarrow}$	0.0195	—	0.0195	0.0195	-0.0306	—	-0.0306	-0.0306	-0.0259	—	-0.0259	-0.0259	0.1064	—	0.1064	0.1064
$\sigma_{\downarrow}$	0.0364	—	0.0364	0.0364	0.1145	—	0.1145	0.1145	0.0778	—	0.0778	0.0778	0.1304	—	0.1304	0.1304
κ	0.0203	—	0.0203	0.0203	-0.0238	—	-0.0238	-0.0238	-0.0226	—	-0.0226	-0.0226	0.1218	—	0.1218	0.1218
R2 Normal																
μ	0.2135	—	0.2135	0.2135	0.1389	—	0.1389	0.1389	0.2188	—	0.2188	0.2188	0.2677	—	0.2677	0.2677
σ	0.1041	—	0.1041	0.1041	0.2411	—	0.2411	0.2411	0.1799	—	0.1799	0.1799	0.2176	—	0.2176	0.2176
λ	3.3245	—	3.3245	3.3245	3.3245	—	3.3245	3.3245	3.3245	—	3.3245	3.3245	3.9894	—	3.9894	3.9894
$\mu_{\downarrow}$	-0.0049	—	-0.0049	-0.0049	-0.0483	—	-0.0483	-0.0483	-0.0169	—	-0.0169	-0.0169	0.0258	—	0.0258	0.0258
$\sigma_{\downarrow}$	0.0259	—	0.0259	0.0259	0.0731	—	0.0731	0.0731	0.0802	—	0.0802	0.0802	0.0728	—	0.0728	0.0728
κ	-0.0045	—	-0.0045	-0.0045	-0.0446	—	-0.0446	-0.0446	-0.0136	—	-0.0136	-0.0136	0.0289	—	0.0289	0.0289
R3 Late-cy																
μ	0.2578	—	0.2578	0.2578	0.2281	—	0.2281	0.2281	0.4313	—	0.4313	0.4313	0.5329	—	0.5329	0.5329
σ	0.0866	—	0.0866	0.0866	0.1723	—	0.1723	0.1723	0.1625	—	0.1625	0.1625	0.1942	—	0.1942	0.1942
λ	5.3333	—	5.3333	5.3333	6.0000	—	6.0000	6.0000	4.0000	—	4.0000	4.0000	2.0000	—	2.0000	2.0000
$\mu_{\downarrow}$	-0.0207	—	-0.0207	-0.0207	0.0164	—	0.0164	0.0164	-0.0075	—	-0.0075	-0.0075	0.0076	—	0.0076	0.0076
$\sigma_{\downarrow}$	0.0208	—	0.0208	0.0208	0.0440	—	0.0440	0.0440	0.0585	—	0.0585	0.0585	0.1010	—	0.1010	0.1010
κ	-0.0203	—	-0.0203	-0.0203	0.0175	—	0.0175	0.0175	-0.0058	—	-0.0058	-0.0058	0.0128	—	0.0128	0.0128
R4 COVID																
μ	0.1032	—	0.1032	0.1032	0.1487	—	0.1487	0.1487	0.2152	—	0.2152	0.2152	0.1095	—	0.1095	0.1095
σ	0.1319	—	0.1319	0.1319	0.2455	—	0.2455	0.2455	0.2236	—	0.2236	0.2236	0.2835	—	0.2835	0.2835
λ	4.6667	—	4.6667	4.6667	4.0000	—	4.0000	4.0000	3.3333	—	3.3333	3.3333	4.0000	—	4.0000	4.0000
$\mu_{\downarrow}$	-0.0107	—	-0.0107	-0.0107	-0.0070	—	-0.0070	-0.0070	0.0170	—	0.0170	0.0170	0.0025	—	0.0025	0.0025
$\sigma_{\downarrow}$	0.0356	—	0.0356	0.0356	0.0791	—	0.0791	0.0791	0.0664	—	0.0664	0.0664	0.0802	—	0.0802	0.0802
κ	-0.0100	—	-0.0100	-0.0100	-0.0039	—	-0.0039	-0.0039	0.0194	—	0.0194	0.0194	0.0057	—	0.0057	0.0057

Notes: This table reports Mean, SD, Min, and Max of the none Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ , σ , λ , $\mu_{\downarrow}$, $\sigma_{\downarrow}$, κ . Each company×regime cell is one estimate: 18-month lookahead ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P6 · GARCH-Merton · none parameter estimates

1.5-year lookback · rolling=none · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ	-0.0305	—	-0.0305	-0.0305	0.6989	—	0.6989	0.6989	-0.0817	—	-0.0817	-0.0817	0.5326	—	0.5326	0.5326
ω	2.695e-06	—	2.695e-06	2.695e-06	4.710e-05	—	4.710e-05	4.710e-05	1.204e-05	—	1.204e-05	1.204e-05	9.001e-04	—	9.001e-04	9.001e-04
α	0.0554	—	0.0554	0.0554	0.0987	—	0.0987	0.0987	0.1011	—	0.1011	0.1011	0.1101	—	0.1101	0.1101
β	0.9275	—	0.9275	0.9275	0.8310	—	0.8310	0.8310	0.8678	—	0.8678	0.8678	3.061e-14	—	3.061e-14	3.061e-14
σ_0	0.0140	—	0.0140	0.0140	0.0262	—	0.0262	0.0262	0.0238	—	0.0238	0.0238	0.0312	—	0.0312	0.0312
λ	0.6667	—	0.6667	0.6667	2.0000	—	2.0000	2.0000	2.0000	—	2.0000	2.0000	2.6667	—	2.6667	2.6667
μ_J	-0.0350	—	-0.0350	-0.0350	-0.0851	—	-0.0851	-0.0851	-0.0138	—	-0.0138	-0.0138	0.1666	—	0.1666	0.1666
σ_J	0.0000	—	0.0000	0.0000	0.0251	—	0.0251	0.0251	0.0906	—	0.0906	0.0906	0.0723	—	0.0723	0.0723
κ	-0.0344	—	-0.0344	-0.0344	-0.0813	—	-0.0813	-0.0813	-0.0096	—	-0.0096	-0.0096	0.1844	—	0.1844	0.1844
R2 Normal																
μ	0.2315	—	0.2315	0.2315	0.1421	—	0.1421	0.1421	0.2948	—	0.2948	0.2948	0.4583	—	0.4583	0.4583
ω	1.379e-05	—	1.379e-05	1.379e-05	6.339e-06	—	6.339e-06	6.339e-06	1.340e-04	—	1.340e-04	1.340e-04	4.158e-05	—	4.158e-05	4.158e-05
α	0.1358	—	0.1358	0.1358	0.0382	—	0.0382	0.0382	0.1734	—	0.1734	0.1734	0.1263	—	0.1263	0.1263
β	0.6047	—	0.6047	0.6047	0.9443	—	0.9443	0.9443	0.2481	—	0.2481	0.2481	0.7358	—	0.7358	0.7358
σ_0	0.0061	—	0.0061	0.0061	0.0151	—	0.0151	0.0151	0.0134	—	0.0134	0.0134	0.0154	—	0.0154	0.0154
λ	1.9947	—	1.9947	1.9947	3.9894	—	3.9894	3.9894	2.6596	—	2.6596	2.6596	3.3245	—	3.3245	3.3245
μ_J	-0.0087	—	-0.0087	-0.0087	-0.0319	—	-0.0319	-0.0319	-0.0098	—	-0.0098	-0.0098	0.0427	—	0.0427	0.0427
σ_J	0.0251	—	0.0251	0.0251	0.0767	—	0.0767	0.0767	0.0908	—	0.0908	0.0908	0.0670	—	0.0670	0.0670
κ	-0.0083	—	-0.0083	-0.0083	-0.0285	—	-0.0285	-0.0285	-0.0057	—	-0.0057	-0.0057	0.0460	—	0.0460	0.0460
R3 Late-cy																
μ	0.2143	—	0.2143	0.2143	0.3426	—	0.3426	0.3426	0.4942	—	0.4942	0.4942	0.7010	—	0.7010	0.7010
ω	2.969e-06	—	2.969e-06	2.969e-06	4.492e-05	—	4.492e-05	4.492e-05	4.208e-05	—	4.208e-05	4.208e-05	5.971e-05	—	5.971e-05	5.971e-05
α	0.2082	—	0.2082	0.2082	0.1209	—	0.1209	0.1209	0.2060	—	0.2060	0.2060	0.1350	—	0.1350	0.1350
β	0.7275	—	0.7275	0.7275	0.6152	—	0.6152	0.6152	0.5131	—	0.5131	0.5131	0.6153	—	0.6153	0.6153
σ_0	0.0044	—	0.0044	0.0044	0.0125	—	0.0125	0.0125	0.0096	—	0.0096	0.0096	0.0143	—	0.0143	0.0143
λ	3.3333	—	3.3333	3.3333	3.3333	—	3.3333	3.3333	2.6667	—	2.6667	2.6667	1.3333	—	1.3333	1.3333
μ_J	-0.0241	—	-0.0241	-0.0241	-0.0029	—	-0.0029	-0.0029	0.0169	—	0.0169	0.0169	0.0406	—	0.0406	0.0406
σ_J	0.0107	—	0.0107	0.0107	0.0502	—	0.0502	0.0502	0.0588	—	0.0588	0.0588	0.1181	—	0.1181	0.1181
κ	-0.0238	—	-0.0238	-0.0238	-0.0017	—	-0.0017	-0.0017	0.0188	—	0.0188	0.0188	0.0488	—	0.0488	0.0488
R4 COVID																
μ	0.2136	—	0.2136	0.2136	0.5268	—	0.5268	0.5268	0.3807	—	0.3807	0.3807	0.5059	—	0.5059	0.5059
ω	4.718e-06	—	4.718e-06	4.718e-06	1.819e-05	—	1.819e-05	1.819e-05	2.013e-05	—	2.013e-05	2.013e-05	1.517e-05	—	1.517e-05	1.517e-05
α	0.1684	—	0.1684	0.1684	0.1773	—	0.1773	0.1773	0.1825	—	0.1825	0.1825	0.2090	—	0.2090	0.2090
β	0.7886	—	0.7886	0.7886	0.7917	—	0.7917	0.7917	0.7474	—	0.7474	0.7474	0.7677	—	0.7677	0.7677
σ_0	0.0107	—	0.0107	0.0107	0.0186	—	0.0186	0.0186	0.0143	—	0.0143	0.0143	0.0142	—	0.0142	0.0142
λ	3.3333	—	3.3333	3.3333	5.3333	—	5.3333	5.3333	2.6667	—	2.6667	2.6667	1.3333	—	1.3333	1.3333
μ_J	-0.0292	—	-0.0292	-0.0292	-0.0204	—	-0.0204	-0.0204	-0.0175	—	-0.0175	-0.0175	-0.0151	—	-0.0151	-0.0151
σ_J	0.0041	—	0.0041	0.0041	0.0666	—	0.0666	0.0666	0.0613	—	0.0613	0.0613	0.0684	—	0.0684	0.0684
κ	-0.0288	—	-0.0288	-0.0288	-0.0180	—	-0.0180	-0.0180	-0.0155	—	-0.0155	-0.0155	-0.0127	—	-0.0127	-0.0127

Notes: This table reports Mean, SD, Min, and Max of the none GARCH-Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ , ω , α , β , σ_0 , λ , μ_J , σ_J , κ . Each company×regime cell is one estimate: 18-month lookback ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.

Table P7 · Heston-Merton · none parameter estimates

1.5-year lookahead · rolling=none · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT				AMZN			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
μ	-0.0912	—	-0.0912	-0.0912	0.4097	—	0.4097	0.4097	-0.1082	—	-0.1082	-0.1082	0.4477	—	0.4477	0.4477
κ	8.7297	—	8.7297	8.7297	13.5431	—	13.5431	13.5431	6.0045	—	6.0045	6.0045	4.4183	—	4.4183	4.4183
θ	0.0456	—	0.0456	0.0456	0.1433	—	0.1433	0.1433	1.000e-04	—	1.000e-04	1.000e-04	0.1687	—	0.1687	0.1687
ξ	0.7469	—	0.7469	0.7469	1.7276	—	1.7276	1.7276	0.1393	—	0.1393	0.1393	1.4347	—	1.4347	1.4347
ρ	-0.9900	—	-0.9900	-0.9900	-0.4922	—	-0.4922	-0.4922	-0.9900	—	-0.9900	-0.9900	-0.9900	—	-0.9900	-0.9900
v_0	0.0359	—	0.0359	0.0359	0.1541	—	0.1541	0.1541	0.1001	—	0.1001	0.1001	0.1226	—	0.1226	0.1226
λ	2.6667	—	2.6667	2.6667	2.0000	—	2.0000	2.0000	2.6667	—	2.6667	2.6667	4.0000	—	4.0000	4.0000
μ_J	0.0195	—	0.0195	0.0195	-0.0306	—	-0.0306	-0.0306	-0.0259	—	-0.0259	-0.0259	0.1064	—	0.1064	0.1064
σ_J	0.0364	—	0.0364	0.0364	0.1145	—	0.1145	0.1145	0.0778	—	0.0778	0.0778	0.1304	—	0.1304	0.1304
κ_J	0.0203	—	0.0203	0.0203	-0.0238	—	-0.0238	-0.0238	-0.0226	—	-0.0226	-0.0226	0.1218	—	0.1218	0.1218
R2 Normal																
μ	0.1945	—	0.1945	0.1945	-0.0234	—	-0.0234	-0.0234	0.1597	—	0.1597	0.1597	0.3665	—	0.3665	0.3665
κ	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	5.7049	—	5.7049	5.7049
θ	0.0211	—	0.0211	0.0211	0.0912	—	0.0912	0.0912	0.0416	—	0.0416	0.0416	0.2627	—	0.2627	0.2627
ξ	2.3915	—	2.3915	2.3915	1.2054	—	1.2054	1.2054	0.0627	—	0.0627	0.0627	0.5838	—	0.5838	0.5838
ρ	-0.7254	—	-0.7254	-0.7254	0.3663	—	0.3663	0.3663	0.9900	—	0.9900	0.9900	-0.9900	—	-0.9900	-0.9900
v_0	0.0187	—	0.0187	0.0187	0.0320	—	0.0320	0.0320	0.0305	—	0.0305	0.0305	0.0384	—	0.0384	0.0384
λ	3.3245	—	3.3245	3.3245	3.3245	—	3.3245	3.3245	3.3245	—	3.3245	3.3245	3.9894	—	3.9894	3.9894
μ_J	-0.0049	—	-0.0049	-0.0049	-0.0483	—	-0.0483	-0.0483	-0.0169	—	-0.0169	-0.0169	0.0258	—	0.0258	0.0258
σ_J	0.0259	—	0.0259	0.0259	0.0731	—	0.0731	0.0731	0.0802	—	0.0802	0.0802	0.0728	—	0.0728	0.0728
κ_J	-0.0045	—	-0.0045	-0.0045	-0.0446	—	-0.0446	-0.0446	-0.0136	—	-0.0136	-0.0136	0.0289	—	0.0289	0.0289
R3 Late-cy																
μ	0.1419	—	0.1419	0.1419	0.3211	—	0.3211	0.3211	0.3944	—	0.3944	0.3944	0.5438	—	0.5438	0.5438
κ	20.0000	—	20.0000	20.0000	14.3824	—	14.3824	14.3824	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000
θ	0.0063	—	0.0063	0.0063	0.1000	—	0.1000	0.1000	0.0665	—	0.0665	0.0665	0.1835	—	0.1835	0.1835
ξ	1.1057	—	1.1057	1.1057	0.9235	—	0.9235	0.9235	0.8312	—	0.8312	0.8312	2.4242	—	2.4242	2.4242
ρ	-0.2763	—	-0.2763	-0.2763	-0.9900	—	-0.9900	-0.9900	-0.9900	—	-0.9900	-0.9900	-0.7107	—	-0.7107	-0.7107
v_0	0.0046	—	0.0046	0.0046	0.0040	—	0.0040	0.0040	0.0057	—	0.0057	0.0057	0.0221	—	0.0221	0.0221
λ	5.3333	—	5.3333	5.3333	6.0000	—	6.0000	6.0000	4.0000	—	4.0000	4.0000	2.0000	—	2.0000	2.0000
μ_J	-0.0207	—	-0.0207	-0.0207	0.0164	—	0.0164	0.0164	-0.0075	—	-0.0075	-0.0075	0.0076	—	0.0076	0.0076
σ_J	0.0208	—	0.0208	0.0208	0.0440	—	0.0440	0.0440	0.0585	—	0.0585	0.0585	0.1010	—	0.1010	0.1010
κ_J	-0.0203	—	-0.0203	-0.0203	0.0175	—	0.0175	0.0175	-0.0058	—	-0.0058	-0.0058	0.0128	—	0.0128	0.0128
R4 COVID																
μ	0.0514	—	0.0514	0.0514	0.1183	—	0.1183	0.1183	0.2690	—	0.2690	0.2690	0.1177	—	0.1177	0.1177
κ	20.0000	—	20.0000	20.0000	20.0000	—	20.0000	20.0000	15.8258	—	15.8258	15.8258	3.0552	—	3.0552	3.0552
θ	0.0304	—	0.0304	0.0304	0.0769	—	0.0769	0.0769	0.0601	—	0.0601	0.0601	0.5123	—	0.5123	0.5123
ξ	2.7233	—	2.7233	2.7233	1.4037	—	1.4037	1.4037	1.1982	—	1.1982	1.1982	3.7150	—	3.7150	3.7150
ρ	-0.9258	—	-0.9258	-0.9258	-0.9900	—	-0.9900	-0.9900	-0.9900	—	-0.9900	-0.9900	-0.9899	—	-0.9899	-0.9899
v_0	0.0434	—	0.0434	0.0434	0.0791	—	0.0791	0.0791	0.0545	—	0.0545	0.0545	0.0357	—	0.0357	0.0357
λ	4.6667	—	4.6667	4.6667	4.0000	—	4.0000	4.0000	3.3333	—	3.3333	3.3333	4.0000	—	4.0000	4.0000
μ_J	-0.0107	—	-0.0107	-0.0107	-0.0070	—	-0.0070	-0.0070	0.0170	—	0.0170	0.0170	0.0025	—	0.0025	0.0025
σ_J	0.0356	—	0.0356	0.0356	0.0791	—	0.0791	0.0791	0.0664	—	0.0664	0.0664	0.0802	—	0.0802	0.0802
κ_J	-0.0100	—	-0.0100	-0.0100	-0.0039	—	-0.0039	-0.0039	0.0194	—	0.0194	0.0194	0.0057	—	0.0057	0.0057

Notes: This table reports Mean, SD, Min, and Max of the none Heston-Merton estimates for SPY, AAPL, MSFT, AMZN. Parameters are μ , κ , θ , ξ , ρ , v_0 , λ , μ_J , σ_J , κ_J . Each company×regime cell is one estimate: 18-month lookahead ending on the first session of the window, held for the year. n is 1. SD is undefined with a single update (shown as —). This is parameter estimation only; it is not an LSM pricing table.